

FOREMOST AGDM, AB, Canada

WMO: 712250

Lat: 49.4833N

Lon: 111.4861W

Elev: 881

StdP: 91.18

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.2	-24.9	-31.7	0.2	-27.9	-28.2	0.3	-24.8	14.5	3.8	12.9	-1.2	3.7	150	0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	32.7	16.0	30.7	15.6	28.6	15.2	18.3	26.8	17.2	26.7	16.3	25.9	4.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	12.2	21.0	14.0	11.1	19.9	12.9	10.3	18.6	55.3	26.9	51.7	26.7	48.9	26.0	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
13.0	11.3	10.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-33.2	35.7	3.0	2.0	-35.4	37.1	-37.1	38.3	-38.8	39.4	-40.9	40.8		
			-33.4	20.6	2.9	1.8	-35.5	21.9	-37.2	22.9	-38.8	23.9	-41.0	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-6.2	-7.0	-0.2	6.0	11.0	15.2	19.2	18.3	13.0	6.3	-1.1	-6.7		
	DBStd	11.17	8.57	8.54	7.38	4.47	4.50	3.31	3.11	3.60	4.75	5.45	7.62	8.19		
	HDD10.0	2484	503	476	317	133	41	2	0	1	22	138	334	518		
	HDD18.3	4723	761	709	574	371	229	105	27	47	168	374	582	777		
	CDD10.0	923	0	0	2	12	74	157	284	256	113	22	2	0		
	CDD18.3	119	0	0	0	0	3	10	53	44	9	0	0	0		
	CDH23.3	2357	0	0	1	8	96	224	952	815	249	13	0	0		
	CDH26.7	890	0	0	0	1	21	58	389	333	88	1	0	0		
(14) Wind	WSAvg	5.0	5.7	5.0	5.5	5.4	5.1	4.6	4.1	4.1	4.6	5.1	5.4	5.3		
(15) Precipitation	PrecAvg	364	21	16	22	27	49	57	33	39	39	14	18	23		
	PrecMax	624	61	42	52	99	115	141	114	111	180	36	54	55		
	PrecMin	231	3	0	0	0	9	1	0	4	2	0	2	5		
	PrecStd	117	17	11	14	21	30	34	25	29	41	11	14	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.7	18.4	24.2	29.1	31.7	35.2	34.5	32.7	25.3	17.9	11.0		
		MCWB	5.7	4.9	8.4	11.0	14.2	17.1	16.8	15.8	14.8	12.5	8.5	4.5		
	2%	DB	7.9	8.4	15.2	20.8	26.1	28.1	33.0	32.5	29.3	21.1	13.6	7.6		
		MCWB	3.6	3.3	7.1	9.9	12.9	15.8	16.7	15.5	13.8	10.7	6.7	3.4		
	5%	DB	5.9	5.9	12.6	17.8	23.4	25.9	30.9	30.5	26.2	17.9	10.9	5.6		
		MCWB	2.4	2.0	5.7	8.2	11.7	15.1	16.5	15.0	13.0	9.3	5.5	2.0		
	10%	DB	4.4	3.8	9.8	15.0	20.7	23.6	28.7	28.1	23.0	15.1	8.5	3.8		
		MCWB	1.6	1.0	4.2	6.5	10.6	14.0	16.2	14.7	11.9	7.9	3.8	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.8	9.1	11.6	14.8	19.1	20.6	18.1	16.3	13.1	9.6	5.7		
		MCDB	10.4	10.8	16.8	22.1	26.4	26.4	28.5	27.4	25.9	23.0	16.1	9.6		
	2%	WB	4.0	3.6	7.5	10.0	13.4	17.3	18.9	17.0	14.7	11.3	7.3	3.6		
		MCDB	7.6	7.9	14.2	20.0	23.8	25.6	27.1	27.5	26.2	19.7	13.2	7.2		
	5%	WB	2.6	2.2	6.1	8.6	12.4	15.9	17.8	16.2	13.6	9.7	5.7	2.2		
		MCDB	5.5	5.4	12.1	16.8	21.6	24.1	26.8	26.9	24.7	16.8	10.4	5.2		
	10%	WB	1.7	1.1	4.4	7.1	11.3	14.7	16.9	15.4	12.4	8.3	4.1	1.1		
		MCDB	4.3	3.7	9.3	13.8	19.1	22.0	26.6	25.6	22.2	14.4	8.1	3.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.3	11.6	13.3	13.2	13.3	16.3	16.0	14.7	13.0	11.3	10.1		
		MCDBR	10.6	13.1	15.8	18.7	18.3	17.6	20.2	20.9	20.5	17.3	14.2	10.8		
		MCWBR	7.4	8.7	8.9	9.1	7.7	7.4	7.5	7.4	8.2	8.5	8.5	7.4		
		MCDBR	10.2	12.3	15.2	17.4	16.0	16.0	16.9	17.0	18.8	16.2	13.7	10.6		
	5% WB	MCWBR	7.4	8.7	8.9	8.8	7.2	7.4	7.6	6.6	7.9	8.3	8.6	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.267	0.287	0.335	0.353	0.347	0.348	0.368	0.328	0.289	0.261	0.251		
	taud		2.448	2.486	2.485	2.349	2.340	2.409	2.419	2.331	2.420	2.516	2.504	2.431		
	Ebn at Noon		816	903	935	912	905	911	904	866	873	860	813	776		
	Edh at Noon		60	74	88	113	119	112	109	114	94	71	56	53		
(44) All-Sky Solar Radiation	RadAvg		1.29	2.29	3.58	4.94	5.84	6.49	7.07	5.67	4.03	2.52	1.43	1.01		
	RadStd		0.11	0.18	0.26	0.34	0.44	0.38	0.36	0.36	0.33	0.24	0.09	0.08		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page